Counterfactual Analysis

Intervention Feature

Pronoun Usage

Intervention Strength (λ): **0.30**

0 = factual · 1 = strong intervention

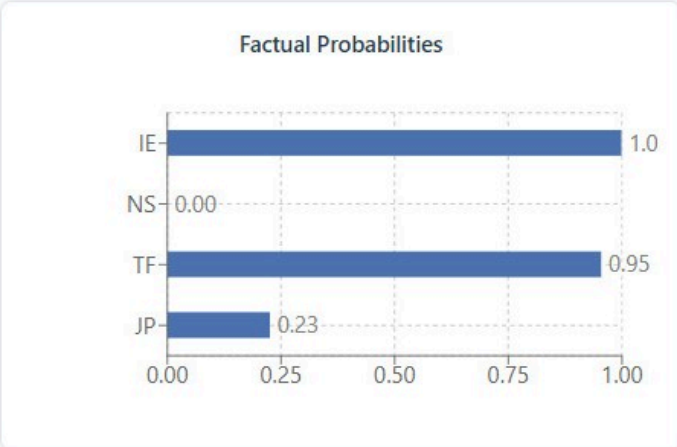
FACTUAL

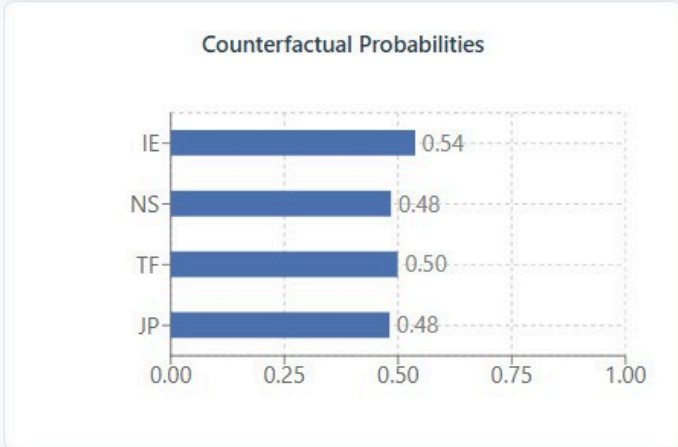
ENFJ

→

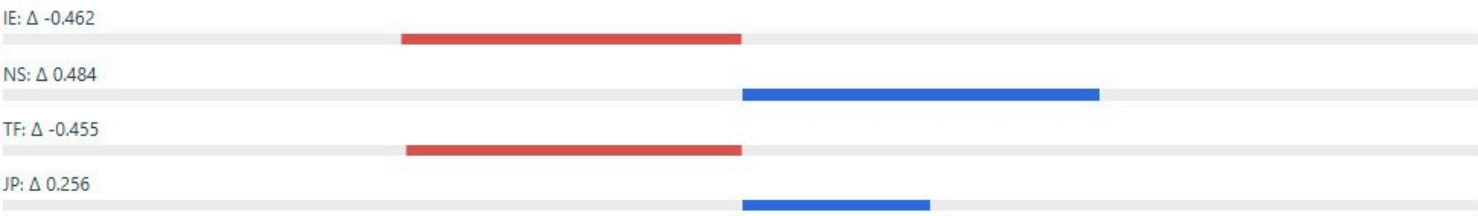
COUNTERFACTUAL

ENTJ





Δ Probability (Causal Shift)



Trait Sensitivity

- IE: 2.671
- NS: 2.802
- TF: 2.632
- JP: 1.478

Decision Logic (Trait Flip)

- **IE**: ✓ Stable
- **NS**: ✓ Stable
- **TF**: ✓ Stable
- **JP**: ✓ Stable

Flip rule: P ≥ 0.6 → ≤ 0.4 (or vice-versa)

Counterfactual Explanation

When intervening on the feature 'pronoun_ratio', the model observed a change of Δ = 0.484 in the NS probability. The sensitivity score for this dimension was 2.802. Although the dominant letter in the NS dimension shifted, the probability change did not cross the calibrated stability boundary. This indicates a marginal shift rather than a robust personality transformation. This reflects the model's design choice to treat personality traits as continuous spectra rather than rigid categories.